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MINIEMO: An Interactive Toy to improve mental health using Generative AI

I. Introduction

AI technology has slowly been making its way in the world as it gets slowly integrated into different aspects of life, however, only recently it has caught widespread public attention. The cause of this are the developments made in the field of generative AI. Whereas traditional AI focuses on detecting patterns, making decisions, honing analytics, classifying data, and detecting fraud, generative AI produces new contents based on input given by the user [1]. In combination with easy-to-use interfaces, generative AI has put AI into the hands of consumers ready and easy to use [2]. This may have created an understanding of the possibilities of AI among the public, leading to hype and many concerns. Generative AI is predicted to remain a big trend in the remainder of 2023 [3] and it offers an opportunity for developers to integrate the technology into consumer products and services.

This fast-paced development in AI is a perfect example of the fast paced, quick changing world we are currently living in [4]. In recent years, the world has been changing ever faster, not only with technological development, but also with political, social, and economical developments. This fast-paced, quick-changing environment has made the world more and more unpredictable, which definitely takes a toll on the general well-being of mankind. Globally mental health conditions are increasing [5, 6], especially among the younger generations [7, 8]. Some of our attention thus has to be spent towards bettering mental health and what better way than using the technology itself to do this.

Therefore, the goal of this project is to create a physical product, with integrated generative AI that has the purpose of improving mental health among young adults and people in their 20s. While working on this project we want to play into everyone's strengths and thus we divided the team up into a technical team and a design team. With the technical team consisting of Soyeong & Selin focusing mainly on the technological aspects (hardware and software) and feasibility. And the design team consisting of Sunah & Vera mainly focusing on the design and cultural aspect of the project.

II. Literature Survey

A way to improve mental health is to create a more positive mindset, by practicing the habit of smiling, meaning that someone makes themselves smile forcefully [9]. Research has shown that when recalling a memory, we use our facial expression to recall the corresponding emotion. So, when we think of a happy memory, we start to smile involuntarily [10]. Moreover, our body uses expressions to reinforce the emotion that we are feeling. When we feel happy, our bodies make us smile to reinforce that emotion [10]. The idea here is that we can use these mechanisms of our body and turn this around. When smiling forcefully, our brain starts to recall and produce happy emotions. Psychologist William James was the first to introduce this idea in the early 19th century [11]. His theory was tested in the 70s by psychologist James Laird and proven true. When forced to smile, people found that they generally felt happier afterwards. This can be explained by the chemical process that takes place in our bodies. When we smile (even forcefully) our body produces three types of hormones that make us feel good namely, dopamine, endorphins and serotonin. This phenomenon has been used in therapy and psychology to help people with mental health problems.

It is thus the goal to remind the user to smile when they have been frowning for too long. The most common methods out there currently are analogue methods of posters and post-it reminders. There are also applications on your phone that simply send the user a reminder to smile randomly during the day. Finally, there is a device developed in China that teaches people how to smile [12].

For this project, it was decided that a doll or avatar of the user that mimics the emotion of the user and consequently reminds the user to smile more was going to be used. The reason for choosing a doll or avatar for this is to create a metaphor. By taking care of the mental health of your avatar, you in turn take care of your own mental health. With this we also want to make use of the Tamagotchi effect which is the development of emotional attachment with machines, robots or software agents [13]. We want the user to care for the doll and to form an emotional attachment with the doll. This will entice the user even more to take care of it and to keep it happy, by remaining a more positive facial expression. Forcefully retaining this positive facial expression will trick the body into being happier. So instead of telling the users to smile more, which can be quite annoying, we want to create intrinsic motivation for the user to smile more on their own.

III. *Material and Method*

1) Art

The design of the avatar-like toy like figure 2, which serves as a representation of the user's appearance, was intentionally devoid of distinctive features. This design approach was motivated by the intention for the device to evoke a sense of personalization akin to a customized toy for any user. This personalized experience is conveyed through the screen embedded in the toy's face. Consequently, other aspects of the device's design were deliberately crafted to be featureless. To adhere to this principle, a monotonous color scheme was chosen. The overall dimensions of the device were set at 30 cm in width, 30 cm in depth, and 25 cm in height, allowing for placement on a desk while ensuring optimal real-time interaction perception.

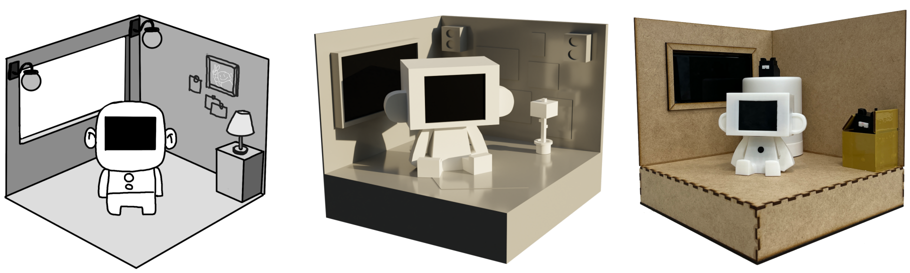

Figure 1. The design process of MINIEMO (Left: concept drawing, Middle: 3D modeling, Right: prototype)

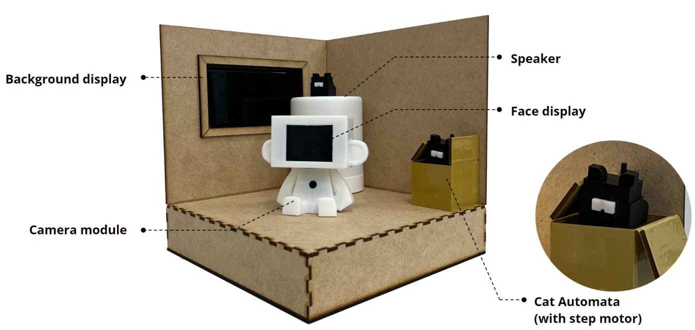

Figure 2. MINIEMO final prototype & components

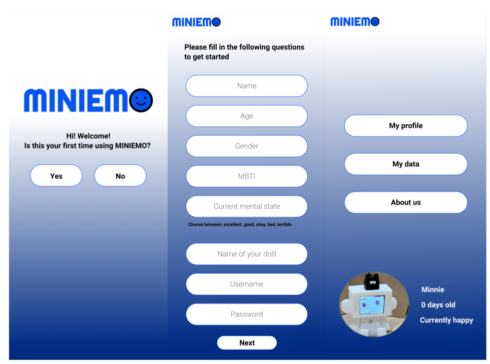

Figure 3. Final prototype of MINIEMO application

2) Technology

In the Technical part, it has various APIs for using Raspberry Pi and Generative AI. There are also various software and hardware that are intimately connected, such as LCD displays and GPIO functions for tact switch control. Those will respond with all physical objects in real-time.

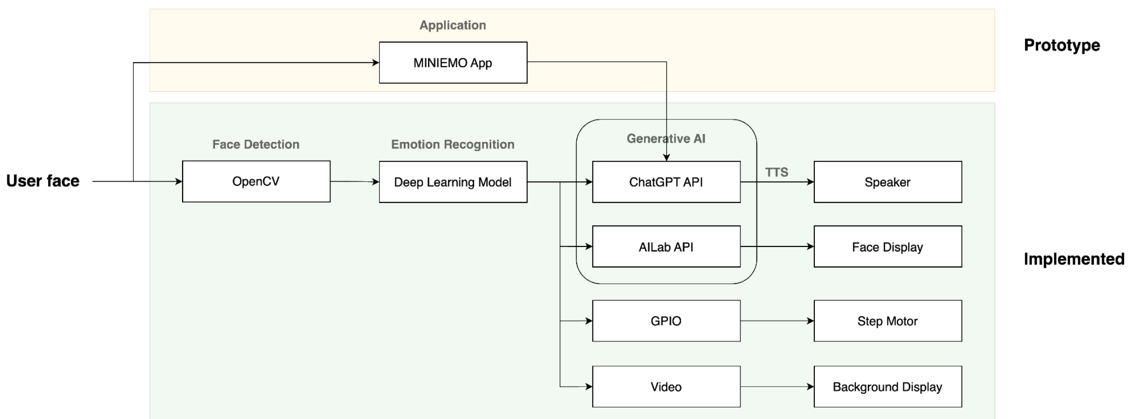

Figure 4. The whole process of MINIEMO

First, the user puts simple information about himself in the MINIEMO application (name, age, gender, MBTI, current mental state etc.). We can check our own data through this application and also check my MINIEMO information.

The following is the core technical part. The camera module connected to the Raspberry Pi recognizes the user's face as OpenCV. When the Haar-Cascade algorithm in OpenCV distinguishes the user's eyes, nose, and mouth, the photo is cropped and put into the input of a deep-learning model that recognizes facial expressions so that the part is well included. The case is divided according to the output expression and the score. If the Sad expression is recognized, ChatGPT automatically generates a sentence that can encourage the user's happy expression according to the user's profile information from the MINIEMO application and the voice is output to the speaker through TTS. If the user's face is recognized as a Happy expression, the cropped user's face image is cartoonized using the AILab API, which is output on Raspberry Pi's

2.2-inch TFT LCD display (model: ILI9341). At the same time, the step motor connected to Raspberry Pi by GPIO rotates 360° and operates cat-shaped automata, adding fun to the user.

According to the seven recognizable facial expressions (happy, sad, natural, angry, discharge, fear, and surprise), the pre-saved video appears as VNC communication on the smartphone screen and is used as a window display. In this case, multi-thread was executed so that the video may be output according to the expression in real-time.

IV. Implementation

We developed an application on a Raspberry Pi 4 using Python 3.7 under the Raspbian OS environment. For camera-related functions, We imported OpenCV. The application takes the user's face as input and uses a trained deep learning model to categorize facial expressions. The threshold for detecting a smiling expression was modified to intensify the detection of smiles.

When the user shows a happy expression, a step motor activates an automaton. Simultaneously, the user's face is cartoonized and displayed on a screen. To ensure that the face and display sizes matched appropriately, We used facial and eye detection techniques. By taking the minimum x-value and the maximum y-value from the bounding box that represents the user's eyes, the application can crop the user's eyes and mouth for the display. If a sad expression is detected, a chatbot powered by ChatGPT offers comforting words, considering the user's inputted name, gender, age, and MBTI personality type. After "please smile" has been prompted twice, the bot generates a message tailored to cheer up the user on the third prompt.

Functions triggered by detecting happy and sad expressions were causing overload on the Raspberry Pi, resulting in system shutdowns. To circumvent this issue, We programmed the camera to deactivate once a certain expression was detected and the corresponding functions were executed. To reactivate the automaton, the user must click a button.

Each time a facial expression is detected, the background video changes. Our original plan was to generate and output a background video using a stable diffusion model, based on inputs such as the user's facial expression, the number of smiles, and MBTI type. However, due to existing latency issues, We instead output pre-generated background videos corresponding to the detected expressions. The camera detection and video playback functions, which run on the main thread, occur concurrently. As a result, the background video can't play while the camera is running. To resolve this, We used multi-threading to enable the video playback. Since a thread, once created, cannot be forcibly terminated until it ends, We limited the background videos to 5 seconds to prevent system overload.

As a result, these functions were able to operate seamlessly in real-time.

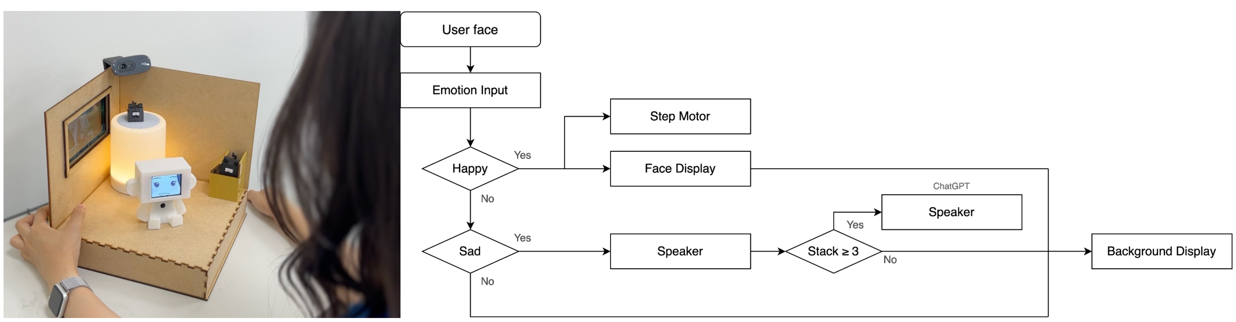

Figure 5. The interaction flow of MINIEMO

V. Demonstration

In order to demonstrate the functionality and features of the device, a demonstration was conducted. The primary focus of this demonstration was to visualize the device's ability to enhance happiness through interactive and personalized experiences. During the demonstration, the connected app, which is linked to the toy, was showcased, and users were prompted to input their personal information. Subsequently, users were seated in front of the activated device and guided to experience the interactions that occur when displaying expressions of sadness and happiness. The objective of this demonstration was to effectively showcase the device's capability to enhance happiness through personalized interactions.

VI. Discussion

For the first, considerable thought was given to the design aspect of MINIEMO for a significant period. The contemplation began from the standard camera shape, pondering over what design could bring joy to the user. Ultimately, by employing a toy shape and cozy space, we believe that MINIEMO can provide a comfortable and familiar experience for users.

Given the amount of visual and auditory information MINIEMO provides to the user, a variety of hardware had to be utilized. Initially, we thought of linking all hardware into a single form factor that could connect to the Raspberry Pi. However, due to issues of additional parts required for connection and stability, some elements were replaced with commercial products. A case in point is the 2.2 inch LCD display, an older model that no longer supports a library that runs a frame buffer copying the HDMI screen from the Raspberry Pi in real-time. This necessitated a considerable amount of time. Although there were a few changes in the final result compared to the prototype, the intention of individual hardware was fulfilled, leading us to conclude that the project was executed successfully in terms of overall completeness.

We also faced several challenges from the software perspective. Originally, we planned to use the Google Cloud API for facial expression recognition. However, calling Google’s API onto a Linux-based single-board computer like the Raspberry Pi and connecting it to the server was not easy, leading to some difficulties. Nonetheless, using a deep-learning model made the process easier. We cloned an existing GitHub of the deep-learning model that recognized expressions and ran it with OpenCV, which was not only more convenient, but also faster and more stable than connecting to the server.

While we didn't conduct a full-scale user study to obtain qualitative and quantitative results, we did present the device to an audience during the final project presentation. Most people responded with broad smiles upon seeing their own reflected facial expression displayed on MINIEMO after they grinned. Additionally, the incorporation of a spatial element resulted in comments about feeling comfortable and relaxed in the space. These reactions suggest that MINIEMO has achieved its goals to some extent.

VII. Conclusion

As previous mentioned, MINIEMO has a Tamagotchi-effect, nurturing a sense of raising a pet in users. The popularity of Tamagotchi can be attributed to the joy of watching a character grow and developing feelings of pride, attachment, and responsibility towards it. Providing users with joy and integrating gaming features such as character development in the app what works with the device can fascinate people. If game-like features are inserted to the device (e.g., accumulating smiles to achieve something), it can serve as motivation for users and offer commercial product potential.

However, what sets MINIEMO apart from other products is the integration of Generative AI withing the device. This allows for not only fast speed and convenient responses, but also user-customized reactions. If a general deep-learning model was embedded in the device, it wouldn't receive updates, and the user would have to continue using the product. But by importing Generative AI in the form of an API, the database accumulates over time and receives responses from their big data. As a result, it can extract only the answers it wants from a vast range of information, particularly the most recent ones. As the device responds in a customized manner to each user's input, it can be permanently used by each individual.

While conducting this project and researching background information, we realized that modern society suffers from serious depression more than expected. According to psychological research, even forced smiles can trick the brain into thinking that it is genuinely smiling, which was new information to us. We decided to address this issue by starting from the intrinsic cause and have put considerable thought into everything from hardware design and composition to the connection method between hardware and software. Effort was also put into the design aspect to make it user-friendly.

If MINIEMO, with its ease of use, friendly features, and potential for generalization, provides more opportunities to smile and maintains a happy mood for longer, it could serve as a solution for alleviating the depression experienced by busy modern people. We believe that this can be one possible approach to solve this societal issue.

VIII. References

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